

7. When we enter a search phrase on the Web search engine, we get a ranked set of results that best match our query, as determined by the search engine. For instance, if you enter the phrase ‘Open Source For You’ in the Google search engine, you get 410,000 results matching it, and the first page of search results contains the ten topmost matches. Given that we have 410,000 results matching the search query, can you come up with an algorithm that does a ranking of these results? Can you identify the typical features that may be used by a search engine to rank its results?
 8. Consider question (7) above. Given a user’s search history across multiple sessions, how will you improve the ranking of search results?
 9. Let us again consider a search engine which accepts a query phrase from the user. Instead of returning all Web pages that match the query phrase, you are asked to design a special search engine that returns only Web pages containing software code that best matches the search phrase. The code search engine takes two inputs — the first is the query phrase delimited using double quotes, and the second input is the programming language for the code results that should be returned by the search engine. For example, if the user enters ‘Insertion Sort’ ‘C++’, then the code search engine should return Web pages that contain the code for ‘insertion sort’ written using the programming language C++.
 - a. For the first question, let us assume that you have a code search engine that can return all Web pages containing code for the search phrase specified. For example, it can return code pages that contain code for insertion sort, written in Python, Java, Scala, Fortran, etc. Now the problem is reduced to that of filtering the code pages that are not in the specified programming language, namely, C++. Can you come up with an algorithm for solving this simplified problem?
 - b. How will you design a code search engine that will return all the Web pages containing code for the search phrase in any programming language?
 10. You are given an array of N records containing employee information, including the employee date of joining, name, address and salary. You are told to sort the array in ascending order in the order of the date of joining of the employee. It is possible that multiple employee records may have the same joining date. In that case, the original order of these records should be retained in the final sorted array. (a) You are told that there is no restriction on the additional space that your algorithm can use. What will your solution be? (b) You are told that you can only use constant additional storage space. What is your solution?
 11. You are given four strings S1, S2, S3 and S4. S1 is a large string and S2, S3, and S4 are comparatively small strings. You are asked to find the smallest length sub-string ‘t’ of S1 such that it contains S2, S3 and S4 as its sub-strings. For example, given string S1 ‘BytheriversofBabylon’, and S2=er, S3=of and S4=so, your function should return ‘ersof’. How will you modify your solution if you are now told that the ordering of S2, S3, and S4 should be maintained in the sub-string ‘t’ (i.e., S2 should appear before S3, and S3 should appear before S4 in sub-string ‘t’)?
 12. In bank cheques, a customer needs to write the amount of the cheque both in the numerical and textual descriptive forms. For instance, if a cheque being issued is for Rs 3238 in numerical form, the customer also needs to provide a textual description of the amount as ‘three thousand two hundred thirty eight’. You are asked to write two functions: (a) function *cheque_itoa*, which takes as input a positive integer and outputs a string that contains the textual description of the input integer; (b) function *cheque_atoi*, which takes as input a string containing the textual description of the cheque amount and returns the numerical integer of this amount.
 13. You are given two sentences S1 and S2, each containing N and M words respectively. You are now asked to convert S1 into S2 by either deleting a word or adding a new word. (Modifying word1 to word2 can be considered as word deletion, followed by word addition.) Each insert operation has a cost of +2 and each delete operation has a cost of +1. Given two arbitrary sentences S1 and S2, the edit distance of (S1, S2) is the minimum total cost of operations needed to convert S1 into S2. You are asked to write a function to compute the edit distance. What is the time and space complexity of your function?
 14. You are given an array of N integers. You are asked to find the largest element which appears an even number of times in the array. What is the time complexity of your algorithm? Can you do this without sorting the entire array?
 15. Given a singly linked list, you are asked to write a function to find the middle element of this list. You are not given the length of the list. How will you modify your solution if the given linked list is a singly linked circular list?
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